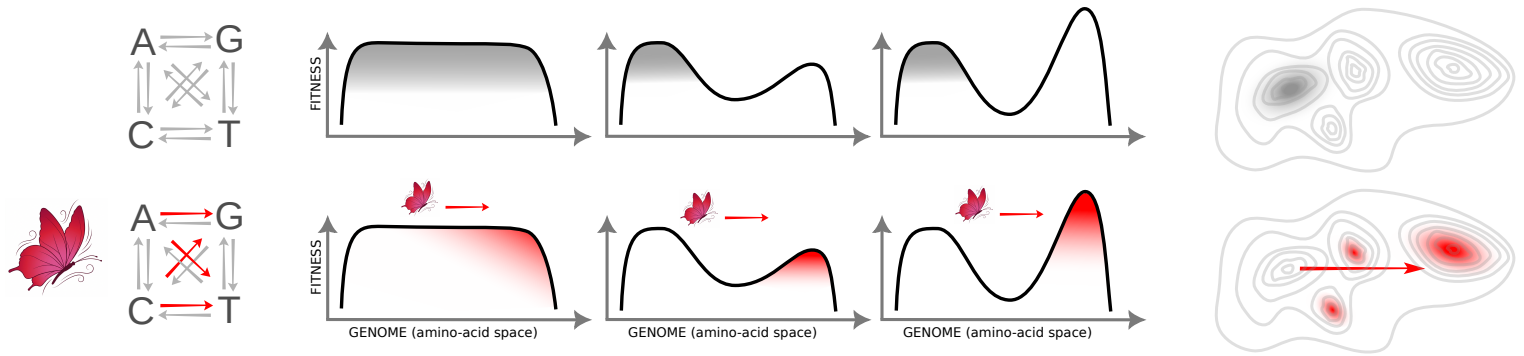


A biological "butterfly effect": small shifts in mutational bias redirect long-term protein evolution



biological state > mutagen exposure > mutational spectrum

genome composition > proteome composition

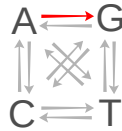
longevity termite whale
non-termite coccroashes mice

temperature/metabolism
coding single-stranded chain
UV exposure..

desamination of ss DNA adenine?



2 types of 12 comp MutSpec



syn codon usage
Gaskew

1st base	2nd base	3rd base	4th base	5th base	6th base	7th base	8th base	9th base	10th base	11th base	12th base
T	A	C	G	A	C	G	A	C	G	A	C
A	T	A	C	T	A	C	T	A	C	T	A
C	T	C	T	C	T	C	T	C	T	C	T
G	C	G	C	G	C	G	C	G	C	G	C
T	C	T	C	T	C	T	C	T	C	T	C
A	C	A	C	A	C	A	C	A	C	A	C
C	G	C	G	C	G	C	G	C	G	C	G
G	A	G	A	G	A	G	A	G	A	G	A
T	A	T	A	T	A	T	A	T	A	T	A
A	C	A	C	A	C	A	C	A	C	A	C
C	T	C	T	C	T	C	T	C	T	C	T
G	C	G	C	G	C	G	C	G	C	G	C

1st base	2nd base											
	T _L A _H	C _L G _H	A _L T _H	G _L C _H	T _L A _H	C _L G _H	A _L T _H	G _L C _H	T _L A _H	C _L G _H	A _L T _H	G _L C _H
Phe	TTT	TCT	TAT	TGT	TTC	TCC	TAC	TGC	TTA	TAA	TAG	TGA
Leu	TTG	TCA	Stop	Trp	TTG	TCG	TAG	TGG	TTA	TAA	TAG	TGA
Leu	CTT	CCT	His	CGT	CTC	CCC	CAC	CGC	CTA	CAA	Arg	CGA
Ile	ATT	ACT	Asn	AGT	ATC	ACC	AAC	AGC	ATA	AAA	Ser	AGA
Met	ATG	ACA	Lys	AGG	ATG	ACG	AAG	AGG	ATA	AAA	Ser	AGA
Val	GTC	GCT	Asp	GGT	GTA	GCC	GAC	GGC	GTG	GAA	Gly	GGA
	GTT	GCG	Glu	GGG	GTT	GCG	GAG	GGG	GTT	GCG	GAG	GGG

кузнечики?
mammals CYTB?

gainer/loser in CYTB (Pro/Phe+LeuTT) ~ generation length (egg size)